

Minseong Kweon

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RESEARCH INTERESTS

3D Reconstruction, Multi-View Geometry, Depth Estimation, SLAM

EDUCATION

University of Minnesota, Twin Cities, United States

Sep. 2025 — present

- M.S. in Robotics

Pusan National University, Busan, Republic of Korea

Mar. 2019 — Feb. 2025

- B.S. in Mechanical Engineering

- Served mandatory military service

Aug. 2020 — May. 2022

UNDER REVIEW

4. **OceanSplat: Object-aware Gaussian Splatting with Trinocular View Consistency for Underwater Scene Reconstruction**

Minseong Kweon, Jinsun Park

- *Under review* (double-blind) → Now in phase 2 review

3. **CAR-Stereo: Confidence-aware Adaptive Disparity Refinement for Real-time Stereo Matching**

Chanill Park, Janghyun Kim, Minseong Kweon, Jinsun Park

- *Under review* (double-blind) → Revise and resubmit

2. **MrGS: Multi-modal Radiance Fields with 3D Gaussian Splatting for RGB-Thermal Novel View Synthesis**

Minseong Kweon, Janghyun Kim, Ukcheol Shin, Jinsun Park (collaboration with CMU RI)

- *Working-in-progress* [\[Project Page\]](#)

* Short version at Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025. [\[Poster\]](#)

* Received **Best Poster Award**.

1. **All-day Depth Completion via Thermal-LiDAR Fusion**

Janghyun Kim, Minseong Kweon, Jinsun Park, Ukcheol Shin (collaboration with CMU RI)

- *Under review* at IEEE Transactions on Intelligent Vehicles (T-IV). [\[arXiv\]](#)

* Short version at Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025. [\[Poster\]](#)

PUBLICATIONS

2. **NoiseGS: Boosting 3D Gaussian Splatting with Positional Noise for Large-Scale Scene Reconstruction**

Minseong Kweon, Kai Cheng, Xuejin Chen, Jinsun Park

- *European Association for Computer Graphics (Eurographics)* short paper, 2025. (Acceptance Rate: 53%) [\[Paper\]](#)

1. **ULTRON: Unifying Local Transformer and Convolution for Large-Scale Image Retrieval**

Minseong Kweon, Jinsun Park

- *Asian Conference on Computer Vision (ACCV)*, 2024. (Acceptance Rate: 32%) [\[Paper\]](#)

AWARDS & HONORS

- **Best Poster Award**, Thermal Infrared Robotics (TIRO) Workshop, ICRA 2025.
- **Honorable Mention**, Undergraduate AI Paper Competition, Summer Annual Conference of IEIE, 2024

ACADEMIC SERVICES

- **Program Committee & Reviewer** AAAI Conference on Artificial Intelligence (AAAI), 2026

WORK EXPERIENCES

Electronics and Telecommunications Research Institute
Research Intern

Jul. 2023 – Aug. 2023
Daejeon, Republic of Korea

Advisor: Dr. Seungmin Choi

Research Area: Visual Localization, Image Retrieval

Project: Visual place recognition for quadruped autonomous robots

- Developed hierarchical visual localization pipeline to achieve robust 6DoF pose estimation for quadruped robots.
- Developed mobile localization application using Unity and NAVER ARC eye API.
- Designed image retrieval module fusing local and global features in visual localization system, reducing the median pose error by 0.27° on the *Aachen Day-Night* dataset without re-ranking.

Related Publication: “ULTRON: Unifying Local Transformer and Convolution for Large-Scale Image Retrieval”, ACCV 2024.

Sandburg Inc.
Software Engineer

Jun. 2022 – Dec. 2022
Busan, Republic of Korea

- Developed a sales-based credit prediction machine learning module for Alternative Credit Scoring System (ACSS).

PRESENTATIONS

3. Self-Similarity Feature Learning for Person Re-Identification

Minseong Kweon, Jinsun Park

- Summer Annual Conference of Institute of Electronics and Information Engineers (IEIE), 2024. [\[Paper \(in Korean\)\]](#)
- * Received **Honorable Mention** at Undergraduate AI Paper Competition.

2. Hierarchical Visual Localization for Indoor Autonomous Systems

Minseong Kweon, Jeonghyun Noh, Chanill Park, Jinsun Park

- Korea Robotics Conferences (KRoC), 2024. [\[Paper \(in Korean\)\]](#)

1. Attention-based Human Activity Recognition with 3-axis Accelerometer Data Conversion

Minseong Kweon, Jaehyeong Park, Kyunghyun Kim, Jeonghyun Noh, Jinsun Park

- Korea Computer Congress (KCC), 2023. [\[Paper \(in Korean\)\]](#)

PATENTS

1. Minseong Kweon, Jinsun Park, “METHOD AND SYSTEM FOR PROCESSING IMAGES BASED ON LOCAL ATTENTION”. Korean Patent Application No. 10-2024-0176234. Patent Pending.

SKILLS

Languages	C/C++, Python, MATLAB, JavaScript
Tools	Github, Linux, Wandb, CUDA programming
Libraries	NumPy, SciPy, Pytorch, Pytorch3D, OpenCV, Open3D, PyBullet, PyRender, scikit-learn